Encryption, ethics, etc.!

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Overview

One-time pads

- Alice and Bob agree on a secret in advance.
- Alice can send one private message to Bob, or Bob to Alice, over public channels.
- Another message to send? Need another secret!

- Upsides: very strong encryption, very fast to encrypt/decrypt, messages go both directions.
- Downsides: can't reuse secrets, need a private channel to arrange secret in the first place.
- Trusting in: the secret is actually secret.

RSA

- Bob generates a *public* key and a *private* key.
- Alice uses the public key to send a message to Bob over public channels.
- Bob decrypts the message with his private key.

- Upsides: Alice can send many messages, no need for a private channel.
- Downsides: could be broken (?), slower than a one-time pad, one directional.
- Trusting in: the difficulty of factoring huge numbers, ownership of the private key.

Exponentiation by Repeated Squaring

Key step in RSA: Bob has private key d and gets encrypted message m^*, wants to compute $\mathrm{rem}((m^*)^d, n)$ to decrypt. But $m*$ is big, and d is big, how to do this quickly?

Can always compute a^k by $k - 1$ multiplications of a.

If k is even, can use $k/2 - 1$ multiplications of a to get $a^{k/2}$. Then, $a^k = (a^{k/2})^2$. So, one more multiplication and we're there.

If k is odd, similar trick to get $a^{(k-1)/2}$, then square, then multiply one more a.

Repeating this idea, the number of multiplications is on the order of $\log k$.

Exponentiation by Repeated Squaring Mod Style

Even better: to compute rem(a^k, n), let's take the mod after each multiplication step!

```
def repsqmodn(a,k,n):
    a := a % n
    if k == 0: return(1)
    if k % 2 == 0:
        sqrootdiva = repsqmodn(a,k/2,n)
        return((sqrootdiva*sqrootdiva) % n)
    sqrootdiva = repsqmodn(a,(k-1)/2,n)
    return((sqrootdiva*sqrootdiva*a) % n)
```

Other RSA uses

As presented: if Bob is the one generating keys, RSA allows Alice to securely message Bob. What else can we do with this encryption scheme?

- Encrypt with Bob's private key. Can be decrypted with Bob's public key. Digital signature—only Bob can make such a message.
- Encrypt with Bob's public key, then Alice's. Bob and Alice must work together to decrypt. Like an encrypted "and."
- Encrypt a one-time pad that can now be sent over public channels.

Breaking RSA

The security of this encryption scheme depends on a “hard problem."

Wiki: In 2019, ... factored a 240-digit number (RSA-240) utilizing approximately 900 core-years of computing power. The researchers estimated that a 1024-bit RSA modulus would take about 500 times as long. (With classical computing techniques.)

Quantum computers can change how this scales. Known ways (Shor’s algorithm) to make this efficient in principle.

So, who can build a real quantum computer?

Musings about algorithms

It’s easy to see algorithms (especially mathematical ones) as being neutral, equalizing.

RSA comes with a mathematical guarantee, so it must be safe, right? Not really.

Important to think of these things—bias, safety, correctness—in context. Is “trust” justified? Who is the adversary?

Closed or open?

COMPAS: a “tool” for predicting likelihood of recidivism for people convicted of crimes.

Notoriously closed-source and unexplained.

Claim: when an algorithm structures lived experience like this, developers have a responsibility for their work.

- Particularly if the algorithm is designed to let others avoid responsibility?

Can we analyze RSA (or encryption more generally) under similar terms? Does the “openness” (and/or simplicity) of RSA help or hinder it?

A Hippocratic Oath for math and CS

(Adapted from `https://arxiv.org/abs/2112.07025` and Alex and Charles)

1. Doctor: I swear to fulfil, to the best of my ability and judgement, this covenant.
   Math/CS: Same.
2. Doctor: I will respect the hard-won scientific gains of those physicians in whose steps I walk, and gladly share such knowledge as is mine with those who are to follow.
   Math/CS: Mathematicians ought to share, publish and disseminate their mathematical knowledge widely.
3. Doctor: I will apply, for the benefit of the sick, all measures [that] are required, avoiding those twin traps of overtreatment and therapeutic nihilism.
   Math/CS: Don't over-model or over-solve problems; don't decide that a problem is unsolveable too early.

A Hippocratic Oath for math and CS

4. Doctor: I will remember that there is art to medicine as well as science, and that warmth, sympathy, and understanding may outweigh the surgeon's knife or the chemist's drug.
   Math/CS: Consider your work in the context of the world, not only as an abstract problem to be solved.
5. Doctor: I will not be ashamed to say "I know not," nor will I fail to call in my colleagues when the skills of another are needed for a patient's recovery.
   Math/CS: There may be times when it is necessary to seek help, guidance, and insight not just from other mathematicians, but also from experts in other fields.
6. Doctor: I will respect the privacy of my patients, for their problems are not disclosed to me that the world may know. Most especially must I tread with care in matters of life and death.
   Math/CS: Respect data privacy; understand that the effects of such data still linger in the trained algorithm long after the initial data has been deleted. Be aware of individual vs group effects.

A Hippocratic Oath for math and CS

7. Doctor: Above all, I must not play at God.
   Math/CS: Same. Don't rely on "objectivity" of math as a source of absolute truth.
8. Doctor: I will remember that I do not treat a fever chart, a cancerous growth, but a sick human being, whose illness may affect the person's family and economic stability.
   Math/CS: An algorithm is a means to an end, not an end in itself.
9. Doctor: I will prevent disease whenever I can, for prevention is preferable to cure.
   Math/CS: Preventing problematic outcomes from our work, in advance, is preferable to the mantra of "moving fast and breaking things."

A Hippocratic Oath for math and CS

10 Doctor: I will remember that I remain a member of society, with special obligations to all my fellow human beings, those sound of mind and body as well as the infirm.
Math/CS: We have all the moral obligations to society that everyone else has, and this is not limited to the entity they work for.

11 Doctor: If I do not violate this oath, may I enjoy life and art, respected while I live and remembered with affection thereafter.
Math/CS: Same!